

AirSense AI

A deep learning based PM2.5 forecasting and weather monitoring application.

Models

- LSTM
- RNN
- GRU

Evaluation

- MAE
- RMSE
- R² Score

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AirSense AI

Deep Learning Based Air Quality Forecasting System

AirSense AI is an intelligent air-quality forecasting system that uses LSTM, RNN and GRU deep learning models to estimate PM2.5 concentration and interpret the predicted air-quality level.

✓ LSTM, RNN and GRU models loaded successfully.

Location & Current Weather

Enter Any Location

Example: Gazipur, Dhaka, Chittagong, London, Dubai...

 Search Weather

Enter any location above to view current weather and its geographical position.

Air Quality & Meteorological Input

Enter the air-quality parameters below. Weather-related parameters are automatically taken from the selected location when weather data is available.

PM10 ($\mu\text{g}/\text{m}^3$)

50.00

- +

O₃ ($\mu\text{g}/\text{m}^3$)

50.00

- +

Dew Point ($^{\circ}\text{C}$)

10.00

- +

SO₂ ($\mu\text{g}/\text{m}^3$)

10.00

- +

Temperature ($^{\circ}\text{C}$)

20.00

- +

Rain (mm)

0.00

- +

NO₂ ($\mu\text{g}/\text{m}^3$)

20.00

- +

Pressure (hPa)

1010.00

- +

Wind Speed (km/h)

2.00

- +

CO

0.50
>  View Current Input Summary



PM2.5 Prediction

 Predict using LSTM

 Predict using RNN

 Predict using GRU



Deep Learning Model Comparison

Compare the PM2.5 predictions produced by LSTM, RNN and GRU using the same input data.

 Compare LSTM vs RNN vs GRU



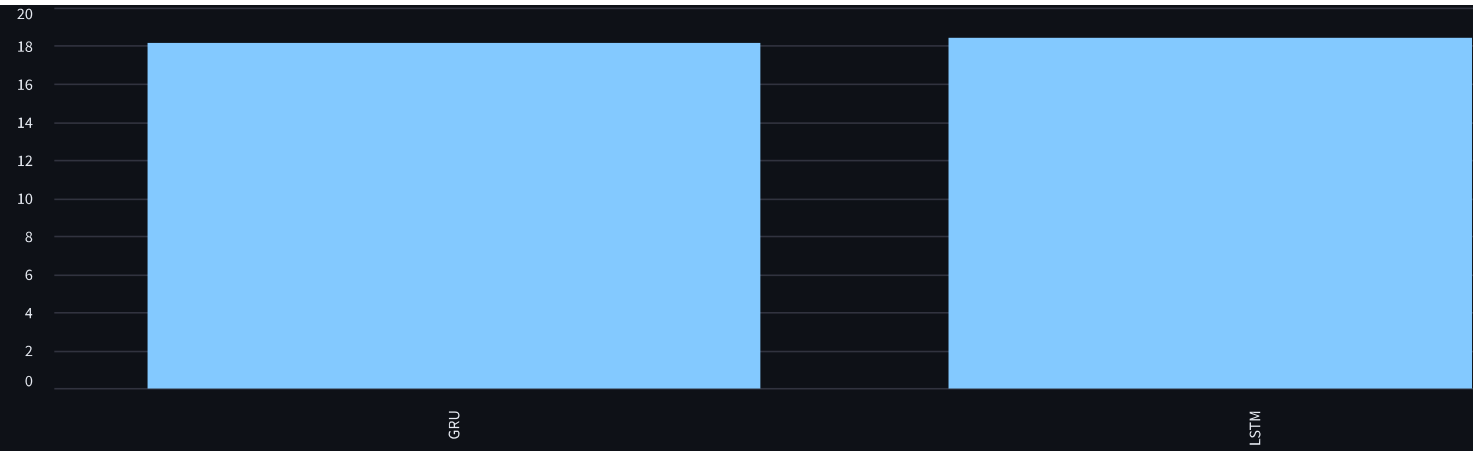
Model Performance

The following evaluation results were obtained from the test data after training the three deep learning models. Lower MAE and RMSE indicate better performance, while a higher R^2 score indicates better performance.

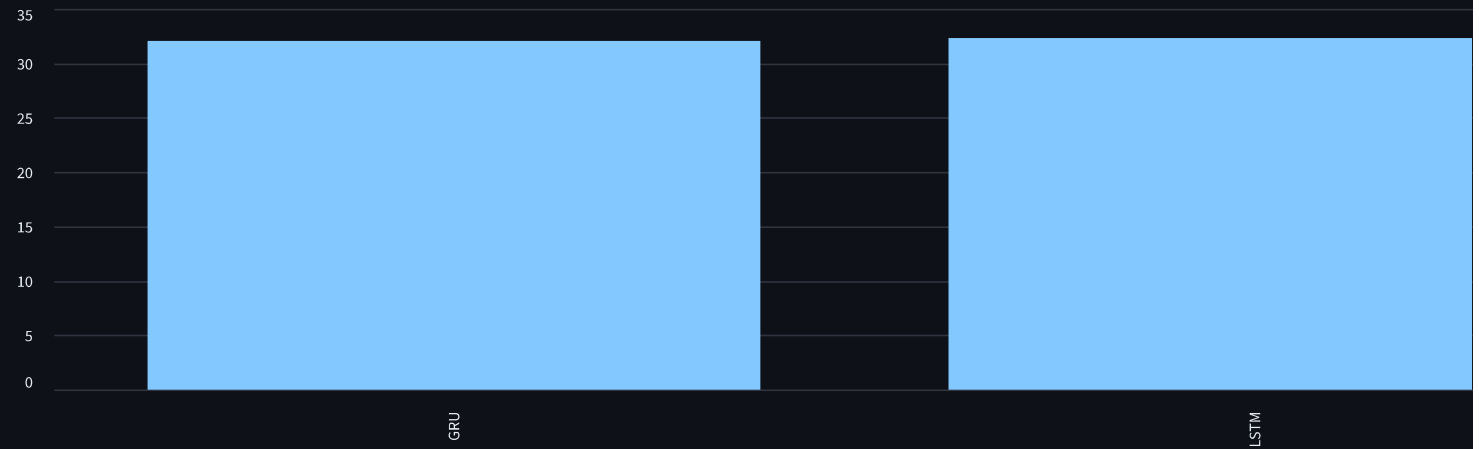
Model	MAE	RMSE
LSTM	18.4159	
RNN	18.9455	
GRU	18.1398	



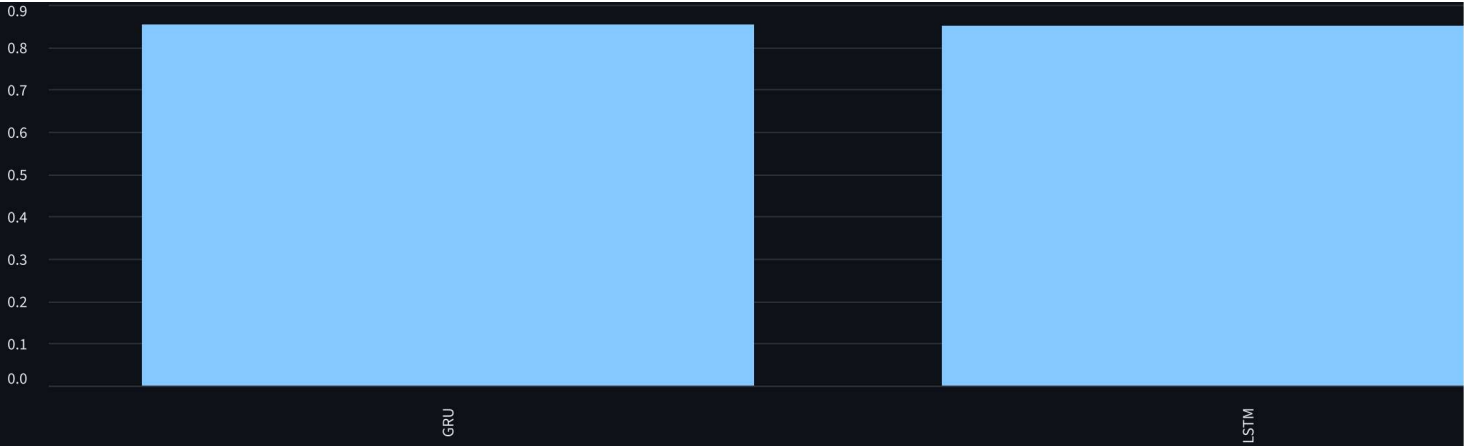
MAE Comparison



RMSE Comparison



R² Score Comparison



🏆 Best Performing Model: GRU

MAE: 18.1398

RMSE: 32.0442

R² Score: 0.8527

GRU achieved the lowest MAE and RMSE and the highest R² score among the three evaluated models.



PM2.5 Air Quality Interpretation

PM2.5 Range (µg/m³)	Air Quality
0 – 9.0	Good
9.1 – 35.4	Moderate / Not Bad
35.5 – 55.4	Unhealthy for Sensitive Groups
55.5 – 125.4	Unhealthy
125.5 – 225.4	Very Unhealthy
> 225.4	Hazardous



Prediction History

No predictions have been made yet.



Download Prediction Report

Make at least one prediction to generate a report.



About AirSense AI

AirSense AI is a deep learning based PM2.5 forecasting system designed to analyze air-quality and meteorological parameters.

The system integrates LSTM, RNN and GRU models for PM2.5 prediction and provides an interpretable air-quality category based on the predicted value.

The application also provides location-based current weather information including temperature, humidity, weather condition, wind speed, pressure, dew point and rainfall.



Input Features

Feature	Purpose
PM10	Air pollutant concentration
SO ₂	Sulfur dioxide concentration
NO ₂	Nitrogen dioxide concentration
CO	Carbon monoxide concentration
O ₃	Ozone concentration
Temperature	Meteorological condition
Pressure	Atmospheric pressure
Dew Point	Moisture-related weather parameter
Rain	Rainfall measurement
Wind Speed	Wind movement

Deep Learning Models Used

Model	Full Name
LSTM	Long Short-Term Memory
RNN	Recurrent Neural Network
GRU	Gated Recurrent Unit

Evaluation Metrics

- MAE — Mean Absolute Error
- RMSE — Root Mean Squared Error
- R^2 — Coefficient of Determination



AirSense AI is ready for location-based weather monitoring, PM2.5 forecasting, model comparison, air-quality interpretation and prediction reporting.